*As of October 18, 2026*

# Chihiro WATANABE

E-mail: chihiro.watanabe.econ@gmail.com
GitHub: https://github.com/Chihiro2000GitHub/
Website: https://Chihiro2000GitHub.github.io/

## Education

**Hitotsubashi University**, Tokyo, Japan *Apr 2026 –*
Ph.D. in Economics

**Hitotsubashi University**, Tokyo, Japan *Apr 2024 – Mar 2026*
M.A. in Economics
Thesis: *"Testing a Macroeconomic Model of Mental Health: Evidence from Micro-Level Causal Inference"*

**Hitotsubashi University**, Tokyo, Japan *Apr 2020 – Mar 2024*
B.A. in Economics, Valedictorian (*Summa Cum Laude*)

## Research Interests

- Health Economics (Mental Health)
- Information Economics (Rational Inattention)
- Computational Psychiatry

## Research

### Work in Progress

- **Rational Delusions** (joint with Hiroki Oi, M.D., Ph.D., psychiatrist and clinical researcher)
  - Initiated an interdisciplinary collaboration integrating economic theory with psychiatric and clinical insights.
  - We develop a rational-inattention model that provides a microfoundation for the "paradox of delusions": the coexistence of two seemingly contradictory features of delusions—jumping to conclusions (JTC), in which weak evidence induces large belief revisions, and bias against disconfirmatory evidence (BADE), in which strong counterevidence induces little revision. In our model, an agent optimally chooses an encoding rule—that is, how to represent and interpret information about the world—by trading off the benefits of accurate information against its processing costs. Beliefs are then updated according to Bayes' rule using the resulting internal representation. Numerical simulations show that the same fixed information-processing cost can generate patterns of both JTC and BADE within a single agent. Our findings suggest that seemingly irrational belief dynamics can emerge from optimal encoding without assuming any impairment in Bayesian inference itself.

**Master's Thesis**

- **Testing a Macroeconomic Model of Mental Health: Evidence from Micro-Level Causal Inference**
  - Replicated in Julia a heterogeneous-agent life-cycle model in which mental health shapes economic decisions through rumination and pessimistically distorted beliefs.
  - Evaluated its quantitative implications using PSID panel data and staggered adoption of state mental health parity laws within a staggered difference-in-differences framework, bridging structural modeling with reduced-form causal inference.

## Research Assistant Experience

- QuantEcon *May 2026 –*
  - Supervisors: Prof. Thomas J. Sargent and Prof. John Stachurski
  - Contribute to maintaining and modernizing open-source materials in computational economics, including NumPy/Numba revisions and textbook review.
- Institute of Economic Research (IER), Hitotsubashi University *Aug 2021 – Mar 2024*
  - Supervisors: Prof. Reiko Gotoh and Prof. Ryo Kambayashi
  - Cleaned and organized survey data for a capability-based study of older adults and persons with disabilities in Kunitachi City.

## Fellowships, Honors & Awards

(HU stands for Hitotsubashi University.)

Selected Participant, RIKEN CBS Brain Science Training Program (BSTP) *2026*
Selected for the Regular Course of the RIKEN Center for Brain Science's training program.

Best Presentation Award, Four-University Joint Research Presentation Meeting, HU *2026*
Selected as the top presentation among 70 posters in the JST SPRING Program.

JST SPRING Fellowship ("The Bridge to the Future" in HU) *2026 – Present*

Moritani Ikueikai Scholarship *2025 – 2026*

Valedictorian, HU *2024*

*Summa Cum Laude* (President's Award), HU Economics *2024*

President's Award (for Outstanding Achievement in a Paper Competition), HU *2024*

Scholarship for Academic Excellence, HU *2023 – 2024*

President's Award (for Highest Academic Performance in Economics, 2022), HU *2023*

## Teaching Experience

- Intermediate Macroeconomics (Master's level) *Fall 2024*
  - Instructor: Prof. Takashi Kano
  - Grading of assignments and exams / Office hours (student support)

## Skills

**Programming:** Julia, Stata, Python, LaTeX
**Languages:** Japanese (native), English
**Tools:** Git, Overleaf, VS Code